

MODULE 5.3

Utility Software & System Maintenance and Security

Utility Software

Programs that help manage, maintain, and control computer resources.



Backup

Creates copies of data to prevent loss.



Disk Scanner

Finds and fixes disk errors.



Defragmentation

Reorganizes files so they open faster.



Compression

Reduces file size (e.g., ZIP).



Task Manager

Monitors running programs and CPU.



Anti-Virus

Protects against malware.

Disk Management

Disk Partitioning

Dividing one physical hard drive into multiple separate logical drives (C:, D:).

Why? Easier to organize OS files separately from user data.

Disk Formatting

Preparing a drive for use by creating a file system.
Erases all data!

Warning: Always Backup data before partitioning or formatting!



Computer Security Threats

Protecting your data from harm.



Malware & Viruses

Harmful software that can delete or steal data.

Solution: Anti-virus



Unauthorized Access

Other people looking at your files.

Solution: Passwords



Physical Damage

Fires, floods, or hardware breaking.

Solution: Backups

Module 5.3 Quick Review

Q: What does Defragmentation do?

A: Reorganizes files to make them load faster.

Q: Why partition a disk?

A: To separate OS files from user data files.

Q: What must you do before Formatting?

A: Backup your data!

Lesson 5 Summary

- ✓ OS links user to hardware.
- ✓ Manages memory, process, devices, files.
- ✓ Files have names and extensions.
- ✓ GUI uses WIMP, CLI uses text.
- ✓ Utilities help maintain the system.
- ✓ Partitioning divides one drive into many.
- ✓ Formatting erases data, needs backup.

End of Lesson 5

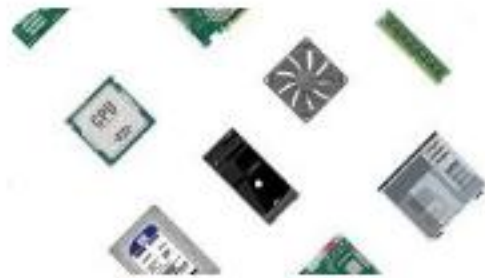
Operating Systems

Image Sources



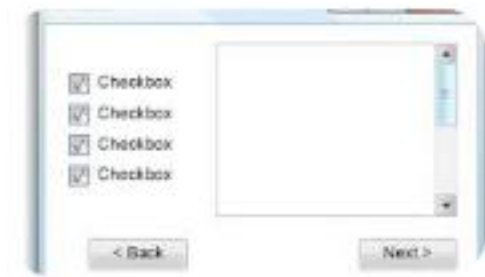
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<https://www.fonedog.com/images/others/partition.png>

Source: www.fonedog.com